Module -7: Network fundamental -

1- Which of the following messages in the DHCP process are broadcasted? (Choose two)

A. Request

B. Offer

C. Discover

D. Acknowledge

Ans: offer and discover

2- Which command would you use to ensure that an ACL does not block web-based TCP traffic?

A. permit any

B. permit tcp any any eq 80

C. permit tcp any eq 80

D. permit any any eq tcp

Ans: permit tcp any any eq 80

3- Explain Network Topologies

Ans: Network topology refers to the physical or logical arrangement of devices in a network.

Bus Topology : All devices share a single cable
Star Topology :Devices connect to a central switch/hub
Ring Topology :Devices form a circular path
Mesh Topology :Every device connects to every other device
Hybrid Topology: Combination of multiple topologies

4- Explain TCP/IP Networking Model

Ans: Layers of TCP/IP Model

Application Layer

Provides services (HTTP, FTP, DNS)

Transport Layer

Handles data delivery ,Protocols: TCP (reliable), UDP (fast)

Internet Layer

Responsible for logical addressing

Uses IP protocol

Network Access Layer
Deals with hardware and physical transmission

5- Explain LAN and WAN Network
Ans: LAN (local area network)
Covers small area
High speed
Eg: Office network

WAN(Wide Area Network)
Covers larger area
Slower than LAN
Eg: Internet

6- Explain Operation of Switch
Ans: Operation of switch are:
A switch operates at layer 2 (data link layer)
Receives data frame
Reads MAC address
Forwards frame to correct port
Reduces collisions
Improves network efficiency

7- Describe the purpose and functions of various network devices
Ans: Various network Devices:

Router: Connects different networks

Switch: Connects devices within LAN

Hub: Broadcasts data to all devices

Firewall: Provides network security

7- Make list of the appropriate media, cables, ports, and connectors to connect switches to other.
Ans:
Cables:

Twisted pair

Switch to pc, switch to router , switch to server

Crossover cable:

Switch to switch, Switch to hub

RJ-45 used with twisted pair cables

8- Define Network devices and hosts

Ans: Network Devices:

Hardware that connects and manages network communication

Examples: Router, Switch, Firewall

Host:

End devices that send or receive data

Examples: Computer, Smartphone, Printer